

Name: _____

This practice exam covers Chapters 1, 2 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Determine whether magnesium and oxygen can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

2. A nurse measures the weight of his patient's medicine. The weight is 0.0084 g. What is the weight in milligrams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

3. Write the chemical formula for phosphoric acid.

ANSWER: _____

4. An optician measures the diameter of her patient's pupil. The diameter is 0.0064 m. What is the diameter in millimeters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

5. Identify whether the following describes a random error or a systematic error, and explain your answer: A balance has not been zeroed before use, so every mass measurement is 0.50 g higher than the true value.

ANSWER: _____

6. Brandon looks up the density of a material and finds it listed as $0.5587 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: _____

7. A botanist measures the thickness of a leaf. The thickness is 4.5 mm. What is the thickness in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

8. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.316 m. What is the thickness in centimeters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

9. As part of her duties as an analytical chemist, Natalie analyzes a sample of an unknown pure substance X and finds that it contains 8.0 g of element E₁ and 2.9 g of element E₂, and no other elements. She then analyzes three other substances that also contain only elements E₁ and E₂ and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E ₁	mass of E ₂
X	8.0 g	2.9 g
A	41.4 g	3.0 g
B	33.4 g	18.9 g
C	33.1 g	6.0 g

ANSWER: _____

10. Hannah must note the concentration of a solution of (NH₄)₃PO₄ in her lab notebook. She knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Hannah weighs an empty graduated cylinder and finds its mass to be 8 g. She then puts some solid (NH₄)₃PO₄ into the graduated cylinder and weighs it. With the (NH₄)₃PO₄ added, the cylinder weighs 25.49 g. She adds water to the graduated cylinder and dissolves the (NH₄)₃PO₄ completely. Then she reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 69.8 mL. What concentration should Hannah write down in her lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

11. Provide the element symbol that represents gallium.

ANSWER: _____

12. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 777 GHz. What is the frequency in hertz (Hz)? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

13. A pharmacist measures the mass of a powdered drug. The mass is 4289 mg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

14. Classify the following as a homogeneous or heterogeneous mixture: a clear sports drink that appears the same throughout when poured. You must explain your answer and use definitions to support your conclusion.

ANSWER: _____

15. A European researcher reports an experimental temperature for a U.S. publication. The temperature is 60°C . Convert this temperature to degrees Fahrenheit and round to the nearest integer.

ANSWER: _____

16. Thomas measures the mass of a copper disc using a balance. The accepted mass of the object is 25.9 g. He records a measurement of 27.14 g. What is the percent error of his measurement? Report your answer to two significant figures.

ANSWER: _____

17. Identify whether the following describes a random error or a systematic error, and explain your answer: Wind gusts cause repeated outdoor temperature measurements to vary randomly around the true value.

ANSWER: _____

18. A botanist measures the thickness of a leaf. The thickness is 1.3 mm. What is the thickness in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

19. Provide the number of protons and the number of neutrons there are in an atom of helium-4.

ANSWER: _____

20. Alexis needs to find the volume of a 13.7 g sample of isopropyl alcohol. The density of the sample is $0.85 \frac{\text{g}}{\text{mL}}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

21. Identify the state of matter in which ethanol exists at 25°C and 1 atm.

ANSWER: _____

22. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: a brass alloy of copper and zinc that appears completely uniform. You must explain your answer with definitions.

ANSWER: _____

23. Provide the element symbol that represents manganese.

ANSWER: _____

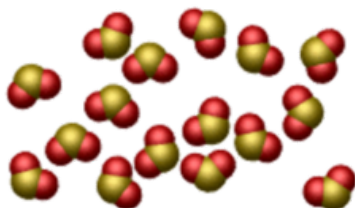
24. A lab technician measures the weight of a reagent. The weight is 2641 g. What is the weight in kilograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

25. State the SI base unit used to measure temperature and give its symbol.

ANSWER: _____

26. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



ANSWER: _____

27. A physical chemist measures the width of a molecule. The width is 58 nm. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

28. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is 8.55×10^{-14} m. What is the wavelength in femtometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

29. Write the chemical formula for chlorous acid.

ANSWER: _____

30. A physical chemist measures the width of a molecule. The width is 79 nm. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

31. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$18.436 \text{ g} - 1.6687 \text{ g}$$

ANSWER: _____

32. Classify the following as a pure substance or a mixture: a handful of garden soil. You must explain your answer and include definitions.

ANSWER: _____

33. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: a bar of pure silver. You must explain your answer with definitions.

ANSWER: _____

34. Write the chemical formula for hydrochloric acid.

ANSWER: _____

35. A chemist notes that a reaction is carried out at -22°F . Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

36. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$40 \frac{\text{mol}}{\text{L}} \times 39. \text{L}$$

ANSWER: _____

37. As part of his duties as an analytical chemist, David measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. He then records these masses in a table in his lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that David would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	16.8 g	9.3 g
2	10.2 g	5.7 g
3	5.7 g	3.2 g
4	7.3 g	4.1 g

ANSWER: _____

38. A shipwright measures the length of a bolt. The length is 77.0 cm. What is the length in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

39. Patricia measures the density of a saline solution in her lab. She finds that the sample has a mass of 127 g and a volume of 36.8 mL. What is the density of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

40. A shipwright measures the length of a bolt. The length is 61.2 cm. What is the length in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

41. Provide the chemical formula for the molecular compound sulfur dichloride.

ANSWER: _____

42. Provide the name and empirical formula of an ionic compound that is formed by the barium cation and the thiocyanate anion.

ANSWER: _____

43. Identify whether the following describes a random error or a systematic error, and explain your answer: A thermometer is poorly calibrated and reads 2.0 °C too high at every temperature.

ANSWER: _____

44. A cell biologist measures the width of a cell. The width is 21 μm . What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

45. Determine whether chlorine and fluorine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

46. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$20.38\text{ mL} - 0.855\text{ mL}$$

ANSWER: _____

47. Classify the following as an element or a compound: pure methane gas. You must explain your answer using correct reasoning.

ANSWER: _____

48. Provide the name for the molecular compound Br_2O_7 . If the compound has a common name, you may provide that instead.

ANSWER: _____

49. An optician measures the diameter of her patient's pupil. The diameter is 0.0015 m. What is the diameter in millimeters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

50. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$200 \text{ mol} \div 23. \text{ L}$$

ANSWER: _____

51. Brian is comparing two unknown liquid samples. Sample A has a mass of 11.8 g and a volume of 28 mL. Sample B has a mass of 11.0 g and a volume of 16 mL. Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

52. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is $9.29 \times 10^{-14} \text{ m}$. What is the wavelength in femtometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

53. A botanist measures the thickness of a leaf. The thickness is 5.8 mm. What is the thickness in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

54. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is $5.84 \times 10^{-14} \text{ m}$. What is the wavelength in femtometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

55. Provide the number of protons and the number of neutrons there are in an atom of argon-40.

ANSWER: _____

56. A nurse measures the weight of his patient's medicine. The weight is 0.0064 g. What is the weight in milligrams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

57. Jonathan measures the mass of a copper disc using a balance. The accepted mass of the object is 35.4 g. He records a measurement of 36.57 g. What is the percent error of his measurement? Report your answer to two significant figures.

ANSWER: _____

58. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$19.422 \text{ g} + 0.8192 \text{ g}$$

ANSWER: _____

59. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$900 \text{ mol} \div 47. \text{ L}$$

ANSWER: _____

60. A nuclear chemist measures the mass of an artificial element. The mass is 34 pg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

61. A geophysicist measures the pressure in a borehole. The pressure is 9.804×10^7 Pa. What is the pressure in megapascals? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

62. Answer the following questions: is Si an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: _____

63. A European researcher reports an experimental temperature for a U.S. publication. The temperature is 50 °C. Convert this temperature to degrees Fahrenheit and round to the nearest integer.

ANSWER: _____

64. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is 7.93×10^{10} m. What is the distance in gigameters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

65. Ian needs to determine the mass of a sample of acetone. The density of the sample is $0.67 \frac{\text{g}}{\text{mL}}$ and its volume is 45.89 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

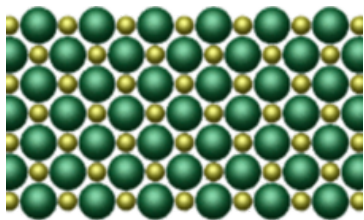
66. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 93 fs. What is the time in seconds? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

67. Isabella needs to know the mass of a sample of $(\text{H}_3\text{O})_3\text{PO}_4$ to 2 significant figures. She puts the sample on a digital scale. This is what the scale shows: 36.9464 g. If this measurement is precise enough for Isabella, round it to 2 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

68. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



ANSWER: _____

69. Does the following process produce a new substance? Salt crystals are stirred into water until they dissolve completely. You must explain your answer with valid reasoning.

ANSWER: _____

70. Provide the element symbol that represents beryllium.

ANSWER: _____

71. Provide the name of the element that the element symbol Kr represents.

ANSWER: _____

72. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is 8.6645×10^{-7} m. What is the wavelength in nanometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

73. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is 8.37×10^{10} m. What is the distance in gigameters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

74. Write the chemical formula of barium oxide tetrahydrate.

ANSWER: _____

75. Provide the name and empirical formula of an ionic compound that is formed by the iron (III) cation and the oxalate anion.

ANSWER: _____

76. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$6.512 \text{ mL} + 0.1190 \text{ mL}$$

ANSWER: _____

77. Olivia needs to find the volume of a 160g sample of isopropyl alcohol. The density of the sample is $0.5639 \frac{\text{g}}{\text{mL}}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

78. Provide the chemical formula for the molecular compound carbon dioxide.

ANSWER: _____

79. A geologist measures the weight of an ore sample. The weight is 5.8364 kg. What is the weight in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

80. Provide the name for the molecular compound Br_2O . If the compound has a common name, you may provide that instead.

ANSWER: _____

81. A millworker measures the width of a plank. The width is 5.0 dm. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

82. Joshua is a thermodynamics researcher who must express a temperature in SI units. He measured the temperature using a Fahrenheit thermometer. The temperature is 68°F . Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: _____

83. State the SI base unit used to measure time and give its symbol.

ANSWER: _____

84. Classify the following as a homogeneous or heterogeneous mixture: a clear sports drink that appears the same throughout when poured. You must explain your answer and use definitions to support your conclusion.

ANSWER: _____

85. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$29. \text{mL} + 3.370 \text{mL}$$

ANSWER: _____

86. Provide the chemical symbol of the most commonly formed ion of iron and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

87. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.242 m. What is the thickness in centimeters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

88. A measurement is reported in kilograms (kg). How many grams is 1 kg equal to?

ANSWER: _____

89. An engineer measures the capacity of an electricity-generating plant. The capacity is 281 MW. What is the capacity in watts? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

90. Classify the reactivity with acid of zinc as a physical property or a chemical property. Explain your answer.

ANSWER: _____

91. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 8.19×10^{-5} g. What is the mass in micrograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

92. A lab technician measures the weight of a reagent. The weight is 6767 g. What is the weight in kilograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

93. Determine whether potassium and chlorine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

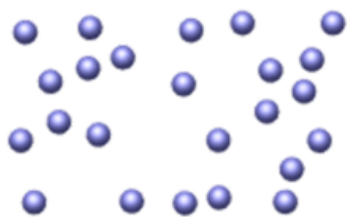
94. A nurse measures the weight of his patient's medicine. The weight is 0.0049 g. What is the weight in milligrams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

95. An engineer measures the capacity of an electricity-generating plant. The capacity is 821 MW. What is the capacity in watts? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

96. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



ANSWER: _____

97. Joshua must note the concentration of a solution of ZnO in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Joshua weighs an empty graduated cylinder and finds its mass to be 5.7 g. He then puts some solid ZnO into the graduated cylinder and weighs it. With the ZnO added, the cylinder weighs 19.38 g. He adds water to the graduated cylinder and dissolves the ZnO completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 41.674 mL. What concentration should Joshua write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

98. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 900 GHz. What is the frequency in hertz (Hz)? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

99. Katherine looks up the density of a material and finds it listed as $2.72 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: _____

100. A geophysicist measures the pressure in a borehole. The pressure is 7.695×10^7 Pa. What is the pressure in megapascals? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

101. A pharmacist measures the mass of a powdered drug. The mass is 6474 mg. What is the mass in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

102. A nurse in a U.S. hospital records a patient's temperature as 68 °F. Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

103. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 232 GHz. What is the frequency in hertz (Hz)? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

104. Provide the name and empirical formula of the ionic compound that is formed by the Cs^+ cation and the F^- anion.

ANSWER: _____

105. Identify the state of matter in which helium exists at 25 °C and 1 atm.

ANSWER: _____

106. A nuclear chemist measures the mass of an artificial element. The mass is 58 pg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

107. Aaron needs to know the mass of a sample of MgI_2 to 2 significant figures. He puts the sample on a digital scale. This is what the scale shows: 90.31 g. If this measurement is precise enough for Aaron, round it to 2 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

108. Classify the following as a homogeneous or heterogeneous mixture: a salad containing lettuce, croutons, and tomatoes. You must explain your answer and use definitions to support your conclusion.

ANSWER: _____

109. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$0.065 \text{ mL} - 0.05569 \text{ mL}$$

ANSWER: _____

110. A civil engineer measures the length of an airport runway. The length is 4527 m. What is the length in kilometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

111. Classify Br_4Si as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

112. State the SI base unit used to measure time and give its symbol.

ANSWER: _____

113. An engineer measures the capacity of an electricity-generating plant. The capacity is 273 MW. What is the capacity in watts? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

114. As part of his duties as an analytical chemist, Shiv measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. He then records these masses in a table in his lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Shiv would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	5.9 g	2.6 g
2	9.6 g	4.3 g
3	6.8 g	3.0 g
4	14.8 g	6.6 g

ANSWER: _____

115. Classify the following as a physical or chemical change: Boiling water on a stove. Explain your answer.

ANSWER: _____

116. Classify the following as a pure substance or a mixture: a diamond gemstone. You must explain your answer and include definitions.

ANSWER: _____

117. Write the chemical formula of zinc fluoride tetrahydrate.

ANSWER: _____

118. Jacob measures the mass of a glass marble using a balance. The accepted mass of the object is 39.7 g. He records a measurement of 37.79 g. What is the percent error of his measurement? Report your answer to two significant figures.

ANSWER: _____

119. William measures the speed of a moving object and records $12 \frac{\text{km}}{\text{h}}$. Convert this speed to $\frac{\text{m}}{\text{s}}$. Ensure your answer has the correct number of significant figures.

ANSWER: _____

120. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 4.28×10^{-5} g. What is the mass in micrograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

121. A millworker measures the width of a plank. The width is 1.0 dm. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

122. Write the chemical formula of silver chromate tetrahydrate.

ANSWER: _____

123. Classify the following as a physical or chemical change: Dissolving table sugar in water. Explain your answer.

ANSWER: _____

124. A millworker measures the width of a plank. The width is 5.1 dm. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

125. Caleb is a thermodynamics researcher who must express a temperature in SI units. He measured the temperature using a Fahrenheit thermometer. The temperature is 86°F . Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: _____

126. A geologist measures the height of a mountain. The height is 5.1786 km. What is the height in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

127. Write the name of the inorganic acid H_3BO_3 .

ANSWER: _____

128. A shipwright measures the length of a bolt. The length is 67.5 cm. What is the length in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

129. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 60 fs. What is the time in seconds? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

130. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is 7.83×10^{10} m. What is the distance in gigameters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

131. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.292 m. What is the thickness in centimeters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

132. A measurement is reported in megagrams (Mg). How many grams is 1 Mg equal to?

ANSWER: _____

133. Write the name of the inorganic acid H_2CO_3 .

ANSWER: _____

134. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$300 \text{ mol} \div 14.2 \text{ L}$$

ANSWER: _____

135. Write the name of the inorganic acid HClO_2 .

ANSWER: _____

136. A physical chemist measures the width of a molecule. The width is 44 nm. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

137. Classify weight as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

138. A pharmacist measures the mass of a powdered drug. The mass is 4340 mg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

139. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is 5.1984×10^{-7} m. What is the wavelength in nanometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

140. A brewer monitors the temperature of a fermentation tank. The temperature of the tank is -4.4°C . Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

ANSWER: _____

141. Classify the following as an element or a compound: solid iron filings. You must explain your answer using correct reasoning.

ANSWER: _____

142. Patricia looks up the density of a material and finds it listed as $3.553 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: _____

143. Answer the following questions: is Zn^{2+} an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: _____

144. Provide the name of the element that the element symbol Sn represents.

ANSWER: _____

145. An optician measures the diameter of her patient's pupil. The diameter is 0.0036 m. What is the diameter in millimeters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

146. Provide the name and empirical formula of an ionic compound that is formed by the calcium cation and the chlorite anion.

ANSWER: _____

147. Allison measures the speed of a moving object and records $38.8 \frac{\text{km}}{\text{h}}$. Convert this speed to $\frac{\text{m}}{\text{s}}$. Ensure your answer has the correct number of significant figures.

ANSWER: _____

148. Classify the following as a pure substance or a mixture: a crystal of pure table sugar (sucrose). You must explain your answer and include definitions.

ANSWER: _____

149. Michael needs to find the volume of a 12.95 g sample of mercury. The density of the sample is $2.422 \frac{\text{g}}{\text{mL}}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

150. Classify the reactivity with acid of zinc as a physical property or a chemical property. Explain your answer.

ANSWER: _____

151. Efe needs to determine the mass of a sample of a saline solution. The density of the sample is $12.4 \frac{\text{g}}{\text{mL}}$ and its volume is 18 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

152. A geologist measures the weight of an ore sample. The weight is 4.2852 kg. What is the weight in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

153. Classify density as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

154. A cell biologist measures the width of a cell. The width is $88 \mu\text{m}$. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

155. A geophysicist measures the pressure in a borehole. The pressure is $7.065 \times 10^7 \text{ Pa}$. What is the pressure in megapascals? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

156. Provide the name and empirical formula of the ionic compound that is formed by the Ca^{2+} cation and the Cl^- anion.

ANSWER: _____

157. Provide the name and empirical formula of the ionic compound that is formed by the Ag^+ cation and the F^- anion.

ANSWER: _____

158. As part of her duties as an analytical chemist, Jennifer analyzes a sample of an unknown pure substance X and finds that it contains 5.3 g of element E_1 and 10.8 g of element E_2 , and no other elements. She then analyzes three other substances that also contain only elements E_1 and E_2 and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E_1	mass of E_2
X	5.3 g	10.8 g
A	8.8 g	3.6 g
B	11.4 g	10.8 g
C	11.1 g	6.9 g

ANSWER: _____

159. Sophia is a thermodynamics researcher who must express a temperature in SI units. She measured the temperature using a Fahrenheit thermometer. The temperature is 59°F . Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: _____

160. David is comparing two unknown liquid samples. Sample A has a mass of 50. g and a volume of 15 mL. Sample B has a mass of 53 g and a volume of 5.29 mL. Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

161. Classify the following as a physical or chemical change: An antacid tablet fizzing in water. Explain your answer.

ANSWER: _____

162. Classify the toxicity of carbon monoxide as a physical property or a chemical property. Explain your answer.

ANSWER: _____

163. A nurse records a patient's temperature. The thermometer reads 87°C . Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

ANSWER: _____

164. Classify mass as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

165. A nuclear chemist measures the mass of an artificial element. The mass is 72 pg. What is the mass in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

166. Classify the following as an element or a compound: a gold nugget refined to 100% purity. You must explain your answer using correct reasoning.

ANSWER: _____

167. David is comparing two unknown liquid samples. Sample A has a mass of 51 g and a volume of 21.3 mL. Sample B has a mass of 10. g and a volume of 6.70 mL. Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

168. Does the following process produce a new substance? A copper wire is bent into a coil. You must explain your answer with valid reasoning.

ANSWER: _____

169. A recipe instructs a cook to preheat an oven to -31°F . Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

170. Sam measures the density of an oil sample in his lab. He finds that the sample has a mass of 40.3 g and a volume of 11 mL. What is the density of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

171. Sophia measures the speed of a moving object and records $12.3 \frac{\text{km}}{\text{h}}$. Convert this speed to $\frac{\text{m}}{\text{s}}$. Ensure your answer has the correct number of significant figures.

ANSWER: _____

172. A lab technician measures the weight of a reagent. The weight is 6186 g. What is the weight in kilograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

173. A civil engineer measures the length of an airport runway. The length is 3249 m. What is the length in kilometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

174. Nicholas measures the density of a saline solution in his lab. He finds that the sample has a mass of 134.0g and a volume of 12.9mL. What is the density of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

175. A cell biologist measures the width of a cell. The width is 23 μm . What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

176. Caleb needs to know the mass of a sample of Al_2S_3 to 3 significant figures. He puts the sample on a digital scale. This is what the scale shows: 114.72 g. If this measurement is precise enough for Caleb, round it to 3 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

177. Provide the name of the element that the element symbol Pu represents.

ANSWER: _____

178. Provide the number of protons and the number of neutrons there are in an atom of cobalt-59.

ANSWER: _____

179. Classify $(\text{H}_3\text{O})_3\text{PO}_4$ as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

180. A geologist measures the height of a mountain. The height is 1.6806 km. What is the height in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

181. A Canadian meteorologist converts a temperature reading for an American audience. The outdoor temperature is -20°C . Convert this temperature to degrees Fahrenheit and round to the nearest integer.

ANSWER: _____

182. Identify the state of matter in which nitrogen exists at 25°C and 1 atm.

ANSWER: _____

183. Provide the name for the molecular compound CO₂. If the compound has a common name, you may provide that instead.

ANSWER: _____

184. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: pure aspirin (acetylsalicylic acid). You must explain your answer with definitions.

ANSWER: _____

185. A brewer monitors the temperature of a fermentation tank. The temperature of the tank is 70°C. Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

ANSWER: _____

186. Classify Cu(OH)₂ as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

187. Provide the chemical symbol of the most commonly formed ion of lithium and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

188. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$90 \frac{\text{mol}}{\text{L}} \times 10. \text{L}$$

ANSWER: _____

189. Provide the chemical symbol of the most commonly formed ion of silver and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

190. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$26. \frac{\text{mol}}{\text{L}} \times 69. \text{L}$$

ANSWER: _____

191. Answer the following questions: is Ag an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: _____

192. Does the following process produce a new substance? Hydrogen gas and oxygen gas are mixed and ignited. Water vapor forms. You must explain your answer with valid reasoning.

ANSWER: _____

193. Emma must note the concentration of a solution of NH_4Br in her lab notebook. She knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Emma weighs an empty graduated cylinder and finds its mass to be 13 g. She then puts some solid NH_4Br into the graduated cylinder and weighs it. With the NH_4Br added, the cylinder weighs 33.23 g. She adds water to the graduated cylinder and dissolves the NH_4Br completely. Then she reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 18.197 mL. What concentration should Emma write down in her lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

194. A geologist measures the height of a mountain. The height is 7.6831 km. What is the height in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

195. As part of her duties as an analytical chemist, Linda analyzes a sample of an unknown pure substance X and finds that it contains 8.9 g of element E_1 and 8.8 g of element E_2 , and no other elements. She then analyzes three other substances that also contain only elements E_1 and E_2 and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E_1	mass of E_2
X	8.9 g	8.8 g
A	47.6 g	18.7 g
B	12.7 g	6.3 g
C	22.2 g	5.5 g

ANSWER: _____

196. A geologist measures the weight of an ore sample. The weight is 5.7755 kg. What is the weight in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

197. Provide the chemical formula for the molecular compound dinitrogen monoxide.

ANSWER: _____

198. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 17 fs. What is the time in seconds? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

199. What multiplicative factor does the metric prefix 'kilo' (symbol: k) represent?

ANSWER: _____

200. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 6.63×10^{-5} g. What is the mass in micrograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____